Ritvik Singh

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Education

University of California, Irvine — Irvine, CA

Bachelor of Science in Computer Engineering | GPA: 3.83 / 4.00 — *Expected May 2028*

- **Relevant Coursework:** Advanced C Programming, Python Programming and Libraries, Introduction to Digital Systems, Digital Logic Design Lab, Organization of Digital Computers, Drones, Boolean Logic and Discrete Structures, Multivariable Calculus, Differential Equations, Linear Algebra

Experience

Electrical Engineering Intern, Hardware Subteam — August 2026 – Present

ZotBins — *Irvine, CA*

- Audited a 72-component ESP32 control board for smart bins used by 1,000+ students, logging 43 design issues
- Caught 10 critical, board-killing faults in the mixed-signal design before fabrication, including 12V fed into 5.5V-max regulators and a boot-strap conflict that trapped the ESP32 in endless boot loops
- Rerouted servo PWM, load-cell, and ultrasonic signals from a slow I2C expander to native ESP32 GPIO to remove timing jitter, and fixed miswired camera data lines that would have produced corrupted trash images
- Co-authoring a solar research paper, modeling a 117 Wh/day worst-case load that firmware sleep could cut 37%

Member, Micromouse and Open Project Space — September 2025 – Present

IEEE at UC Irvine — *Irvine, CA*

- Building a robot that maps and solves a maze on its own for the All American Micromouse Competition, running floodfill pathfinding on an STM32 that reads 4 IR wall sensors and wheel encoders in real time
- Assembling and bringing up the mouse's STM32 control board, validating 3.3V/5V regulator rails, ideal-diode power protection, crystal clocking, and SWD flashing over ST-LINK before motor integration
- Built 7 embedded projects and a custom-PCB rover capstone spanning C++ firmware, analog circuits, and wireless IoT, applying KiCad schematic capture, SMD rework, and oscilloscope debugging to each build

Projects

F.R.N.D. – Serverless Emergency Drone Network — March 2026

IrvineHacks 2026, Qualcomm Sponsor Track — *Edge AI, Embedded Linux, Python, WebSockets*

- Engineered an offline emergency network in 48 hours with a 4-person team, deploying Raspberry Pi Wi-Fi access points on drones so any phone can reach a first responder with no internet, cell service, app, or account
- Deployed quantized on-device edge AI on an Arduino Uno Q (Qualcomm QRB2210, embedded Linux), running a 4-bit SmolLM2-135M LLM via llama.cpp that triages each caller's location, injuries, and urgency in real time
- Integrated the AI node into an asyncio Python WebSocket relay that routes phone and AI clients by URL path

RC Rover with Custom PCB Remote — April 2026

IEEE Open Project Space Capstone — *Embedded C++, ESP32-C3, KiCad PCB Design, ESP-NOW, PWM*

- Designed and fabricated a 2-layer KiCad PCB for an ESP32-C3 remote streaming joystick data over ESP-NOW
- Achieved 19 of 20 collision-free autonomous runs by writing embedded C++ obstacle-avoidance firmware with ultrasonic ranging, an 8-heading scan for the clearest path, and a 10 ms non-blocking manual override
- Cut straight-line drift 95% (9 cm to 4 mm over 2 m) by trimming L293D H-bridge PWM for mismatched motors

AI-Assisted Portfolio Website — October 2026

Personal Project, ritviksingh.net — *React, JavaScript, Firebase, Cloudflare, Claude*

- Shipped a responsive 7-page React/Vite portfolio in 3 weeks scoring 92+ in all 4 Lighthouse categories, directing Claude (AI-assisted development) against a written spec of design tokens and locked rules across 15 merged PRs
- Developed a JavaScript SVG engine routing circuit traces from live DOM layout, with SPF/DMARC-secured DNS

Skills & Interests

Programming: Python, Java, JavaScript, C, Embedded C++, HTML, Verilog
Hardware & Embedded: ESP32, Arduino, Raspberry Pi, PCB Design, Circuit Design, SMD Soldering, Oscilloscope
Design & CAD: KiCad, SolidWorks, Fusion 360
Web & AI: React, Vite, Firebase, Cloudflare, Edge AI (llama.cpp), AI-Assisted Development
Tools & Protocols: AWS, Git, Linux, MATLAB, MQTT, I2C, SPI, WebSockets, ESP-NOW, PWM, SQL
Certificates: AWS Certified Solutions Architect – Associate